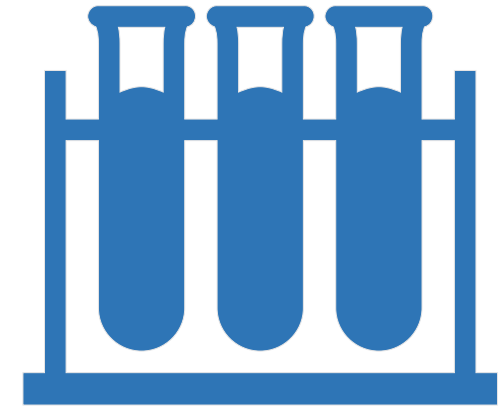


Pt. 2



F-tests, regression,
and ANOVA



Picking up where we left off



This Time...
The idea of analysis of variance

- When we ask if a set of sample means gives evidence for differences among the population means,
 - what matters is not how far apart the sample means are, but
 - how far apart they are relative to the variability of individual observations.



The idea of
analysis of
variance

The idea of analysis of variance

Analysis of variance compares the
variation

due to specific sources with the variation
among individuals who should be similar.

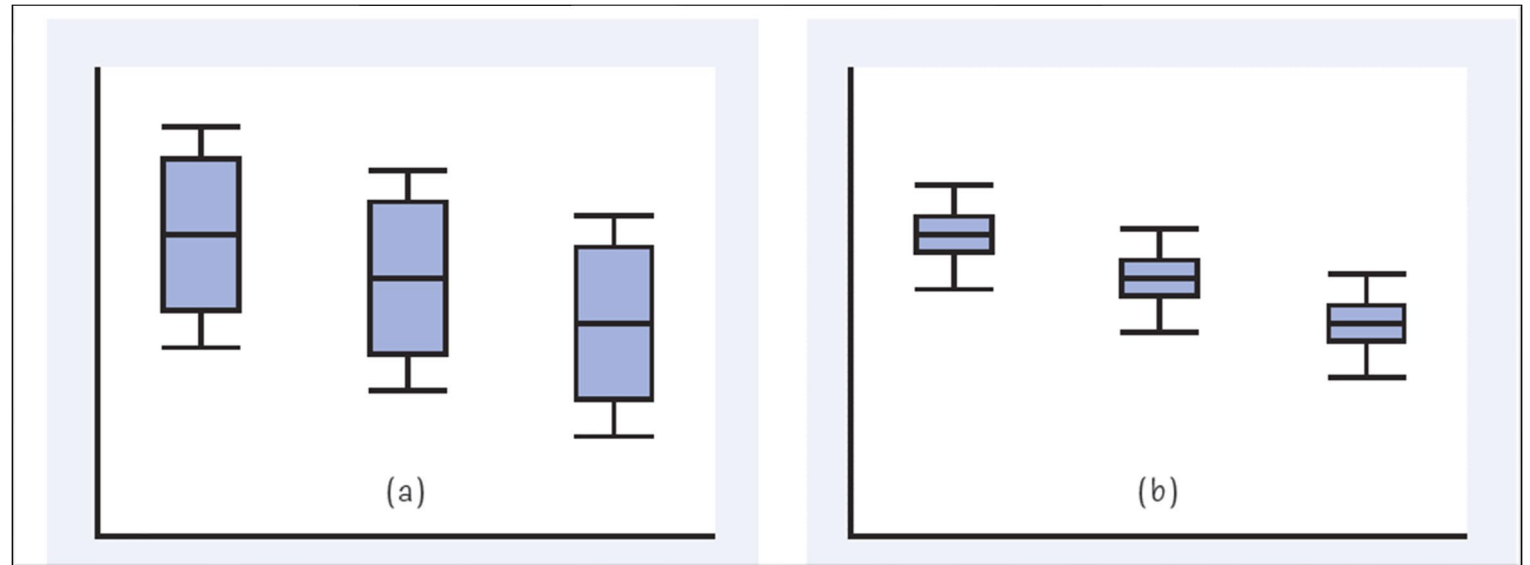


ANOVA tests whether several
populations have the same mean,

by comparing how
far apart the sample
means are

with how much
variation there is
within the sample.

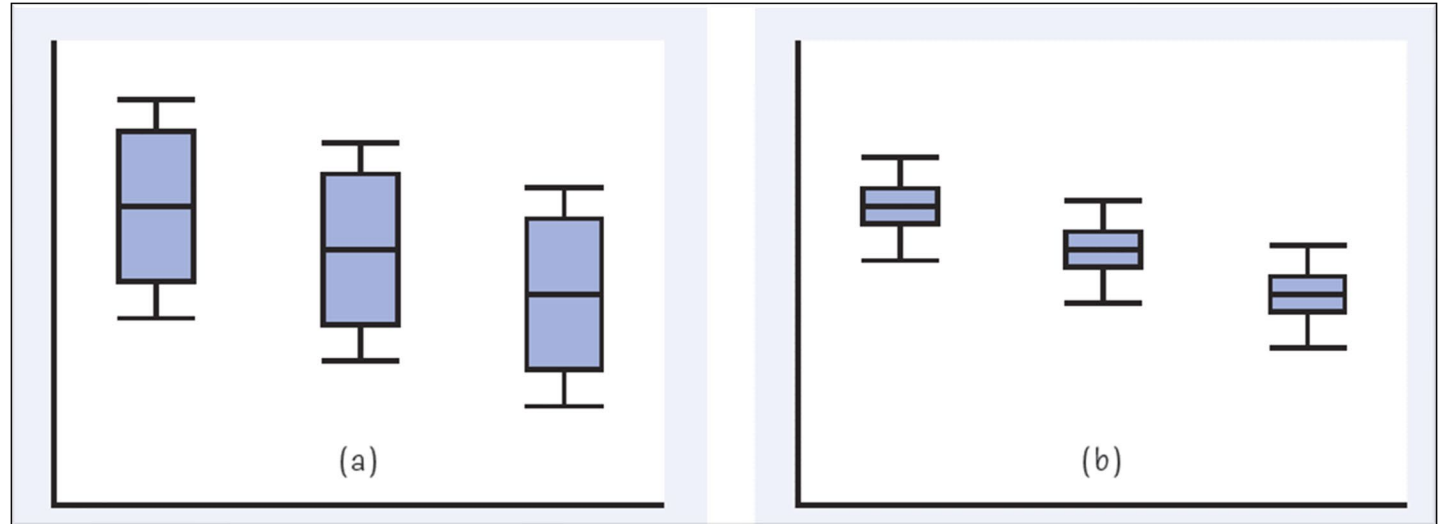
Variation within and between



- The sample means for the 3 samples are the same for each set (a) and (b) below:
 - The variation among sample means for (a) is identical to (b).
 - The variation among the individuals within the three samples is much less for (b).

Interpretation

- The samples in (b) contain a larger amount of variation among the sample means relative to the amount of variation within the samples,
- ANOVA have greater power to differences among the means in (b).
 - Assume equal sample sizes here for (a) and (b).
 - Note: Larger samples have greater power to detect differences.



The ANOVA F statistic



One of the oddities of statistical language is that methods for comparing means are named after the variance. The reason is that the test works by comparing two kinds of variation.



Analysis of variance (ANOVA) is a general method for studying sources of variation in responses.



Comparing several means is the simplest form of ANOVA, called **one-way ANOVA**.

The ANOVA F statistic

- The analysis of variance F statistic for testing the equality of several means has this form:
- $$F = \frac{\text{variation among the sample means}}{\text{variation among individuals in the same sample}}$$
- F must be zero or positive, with F being zero only when all sample means are identical and F getting larger as means move further apart.
- Large values of F are evidence against H_0 : equal means; while the alternative H_a is many-sided, the F test is upper one-sided.

A close-up, shallow depth-of-field photograph of an hourglass. The hourglass is made of clear glass and is filled with a vibrant red liquid. The liquid is captured in mid-pour, creating a thin, continuous stream that falls from the narrow neck into the lower bulb. The background is heavily blurred, showing soft, out-of-focus shapes of other hourglasses, suggesting a collection or a display. The lighting is soft and even, highlighting the transparency of the glass and the rich color of the liquid.

Wrapping Up...

A close-up, shallow depth-of-field photograph of an hourglass. The hourglass is made of clear glass and is filled with a vibrant red sand. The sand is captured in mid-fall, creating a thin, vertical stream that tapers as it approaches the narrow neck. The background is a soft, out-of-focus light gray, with blurred shapes of other hourglasses visible on either side, suggesting a larger collection or a conceptual setting. The lighting is soft and even, highlighting the texture of the sand and the smooth curves of the glass.

Variability

This Time...

The ANOVA F statistic

- The analysis of variance F statistic for testing the equality of several means has this form:
- $$F = \frac{\text{variation among the sample means}}{\text{variation among individuals in the same sample}}$$
- F must be zero or positive, with F being zero only when all sample means are identical and F getting larger as means move further apart.
- Large values of F are evidence against H_0 : equal means; while the alternative H_a is many-sided, the F test is upper one-sided.

Between-Group Variability

- Consider the distributions of the below two samples.
- As these samples overlap,
 - their individual means won't differ by a great margin.
- The difference between their **individual means** and the **grand mean** won't be distinct enough.



Between-Group Variability

- Now consider these two sample distributions:
- As the samples differ from each other by a bigger margin,
 - their individual means would also differ.
- The difference between the
 - individual means and
 - grand mean
 - would therefore also be significant.



Between-Group Variability

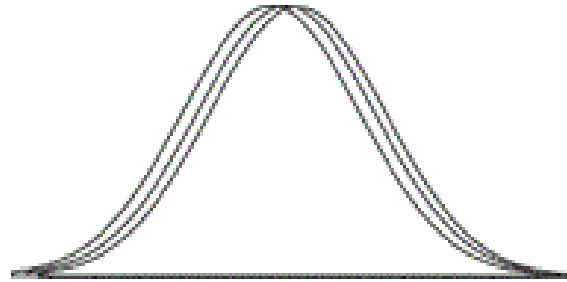
- Such variability between the distributions called **Between-group variability**.
- It refers to variations *between the distributions* of individual groups (or levels) as the values within each group are different.
 - In ANOVA, this variability is captured by **mean square for groups** (MSG)



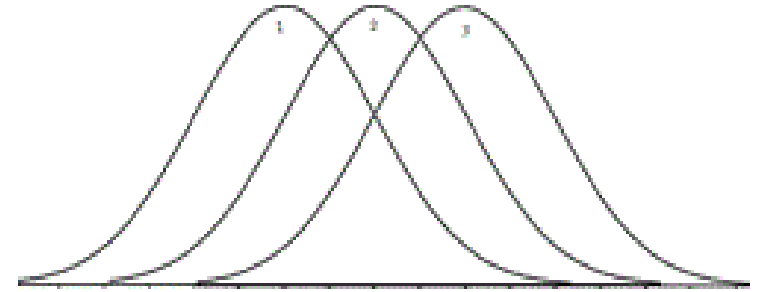
Between-Group Variability

- Each sample is looked at and the difference between its mean and grand mean is calculated to calculate the variability.
- If the distributions overlap or are close,
 - the grand mean will be like the individual means
- whereas if the distributions are far apart,
 - difference between means and grand mean would be large.

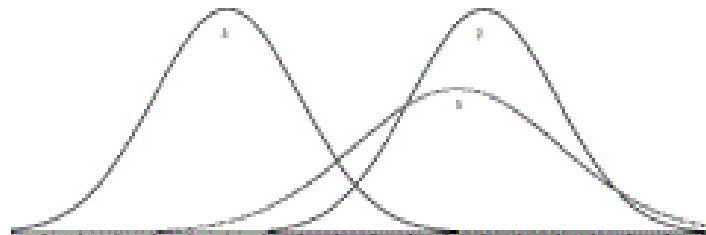
Little discrimination



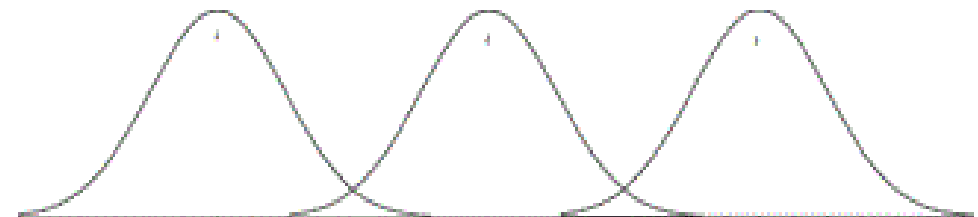
Some Discrimination



Discrimination between Two Groups,
but not the third

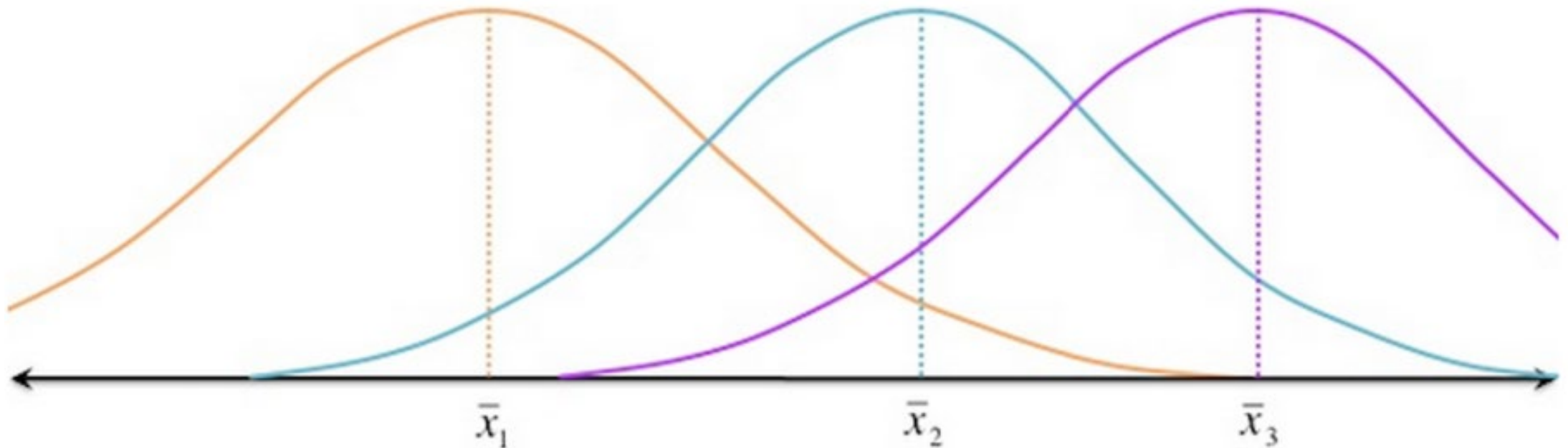


Large Discrimination



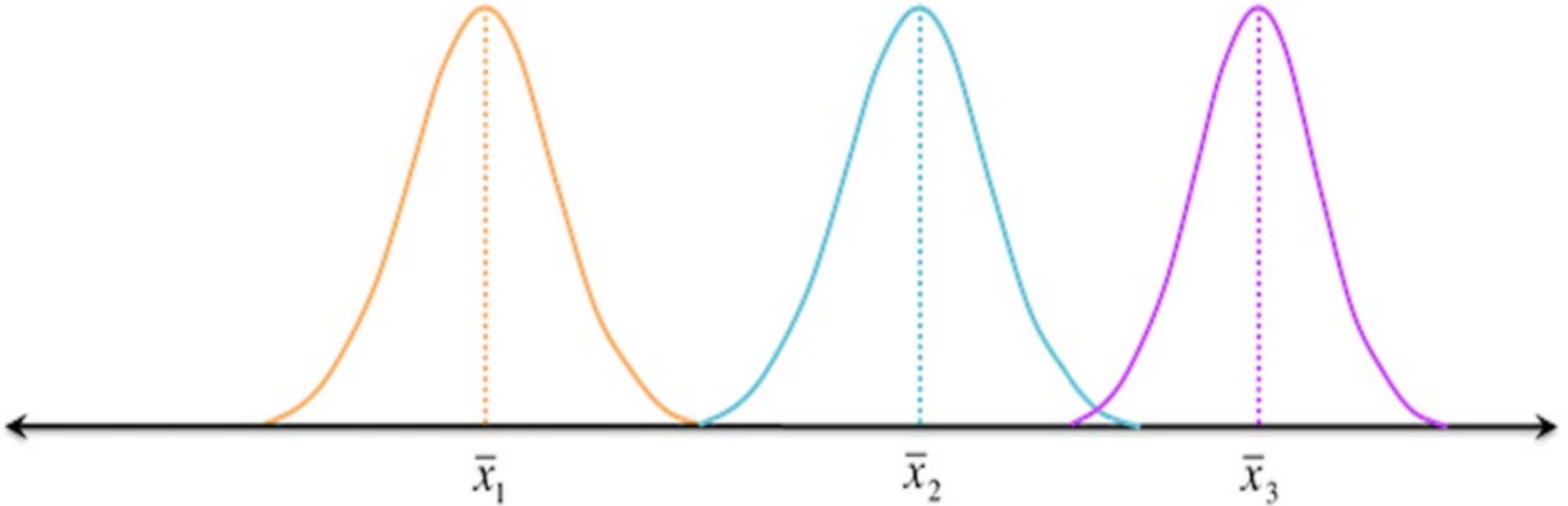
Within-Group Variability

- Consider the given distributions of three samples.
- As the spread (variability) of each sample is increased,
 - their distributions overlap and
 - they become part of a big population.



Within-Group Variability

- Now consider another distribution of the same three samples
 - but with less variability.
- Although the means of samples are similar to the samples in the above image,
 - they seem to belong to different populations.



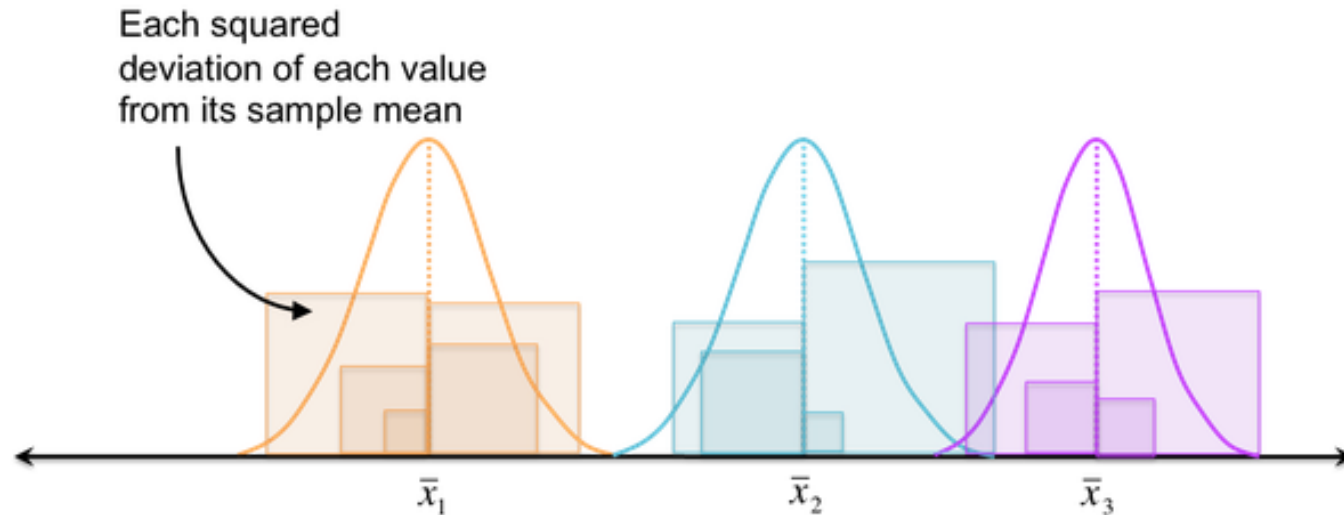
Within-Group Variability

- Such variations within a sample are denoted by **Within-group variation**.
- It refers to variations caused by differences within individual groups (or levels)
 - as not all the values within each group are the same.
- Each sample is looked at on its own
 - and variability between the individual points in the sample is calculated.
 - (In other words, no interactions between samples are considered.)



Within-Group Variability

Note: x_{i1} is the i th value from the first sample, x_{i2} is the i th value from the second sample, and so on all the way to x_{ik} , the i th value from the k th sample. x_{ij} is therefore the i th value from the j th sample.



With within-group variability, SS_{within} is the sum of each squared deviation of each value from its respective sample mean (the total area of all the squares in the figure above). MS_{within} is the average-sized square.

- We can measure **Within-group variability**
 - by looking at how much each value in each sample differs from its respective sample mean.
- So first, we'll take the squared deviation of each value from its respective sample mean and add them up.
- This is the **sum of squares for within-group variability**.

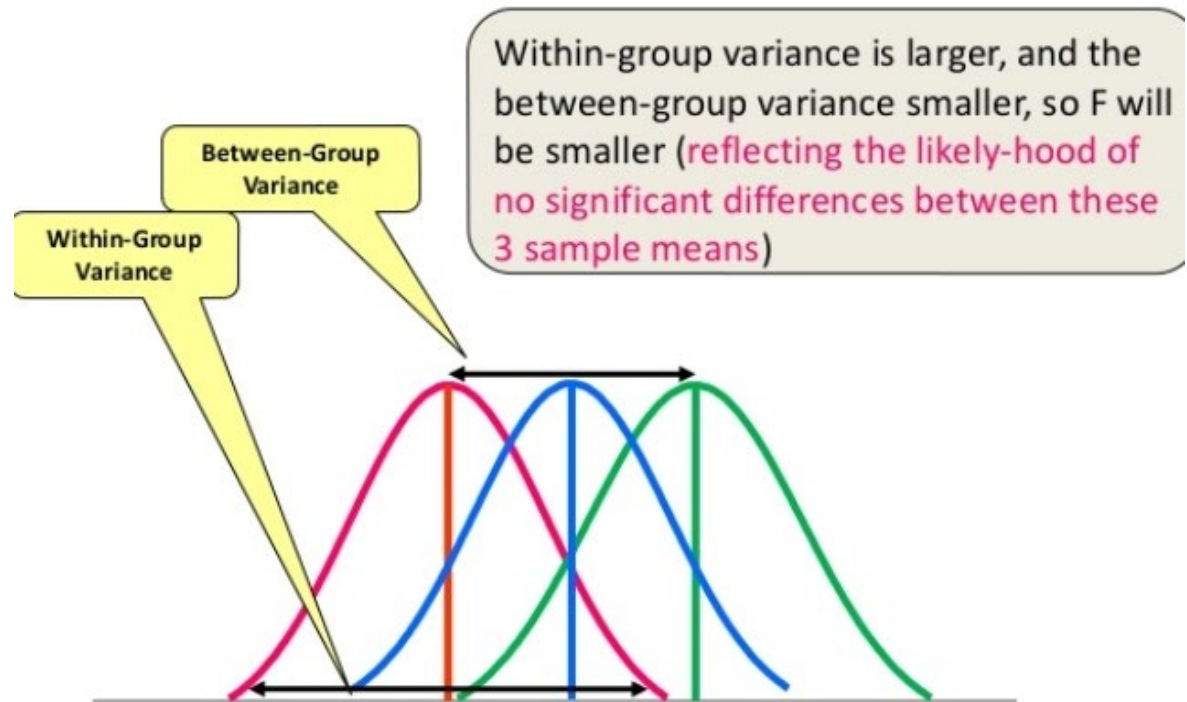
Some details of ANOVA

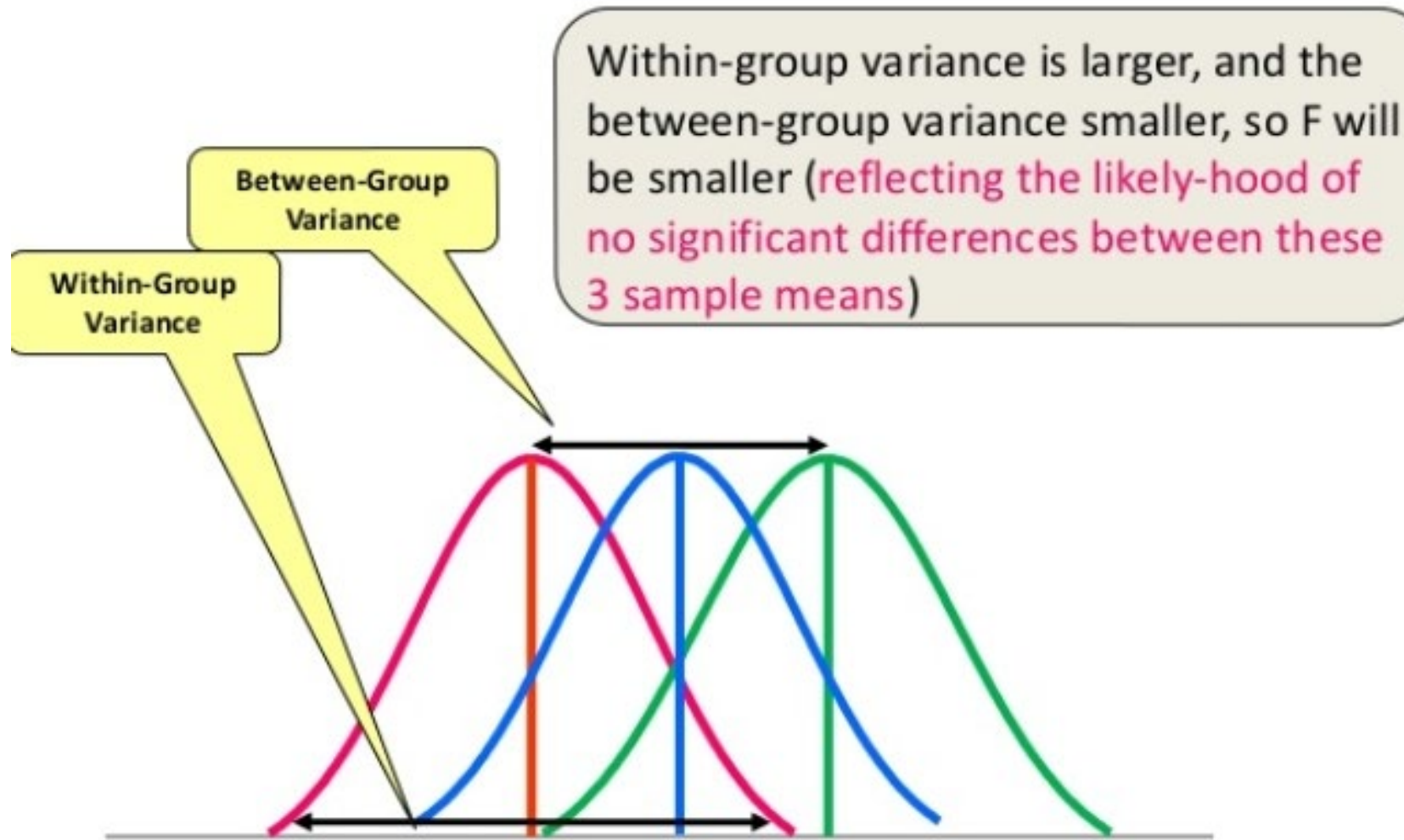
- The mean square in the numerator of F is an average of the squares of these deviations.
- We call it the mean square for groups (MSG):
- $$MSG = \frac{n_1(\bar{x}_1 - \bar{x})^2 + n_2(\bar{x}_2 - \bar{x})^2 + \dots + n_I(\bar{x}_I - \bar{x})^2}{I - 1}$$
- The mean square in the denominator of F measures variation among individual observations in the same sample.
- For all i samples together, we use an average of the individual sample variances.
 - We call it the mean square for error (MSE):
- $$MSE = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 + \dots + (n_I - 1)s_I^2}{N - I}$$

Putting the Pieces together

- Draw an independent SRS from each of i Normal populations that have a common standard deviation but may have different means.
- The sample from the i^{th} population has size n_i , sample mean $\bar{x}_i$, and sample standard deviation s_i .
- To test the null hypothesis that all i populations have the same mean against the alternative hypothesis that not all the means are equal,
 - calculate the **ANOVA F statistic**:

$$F = \frac{MSG}{MSE}$$



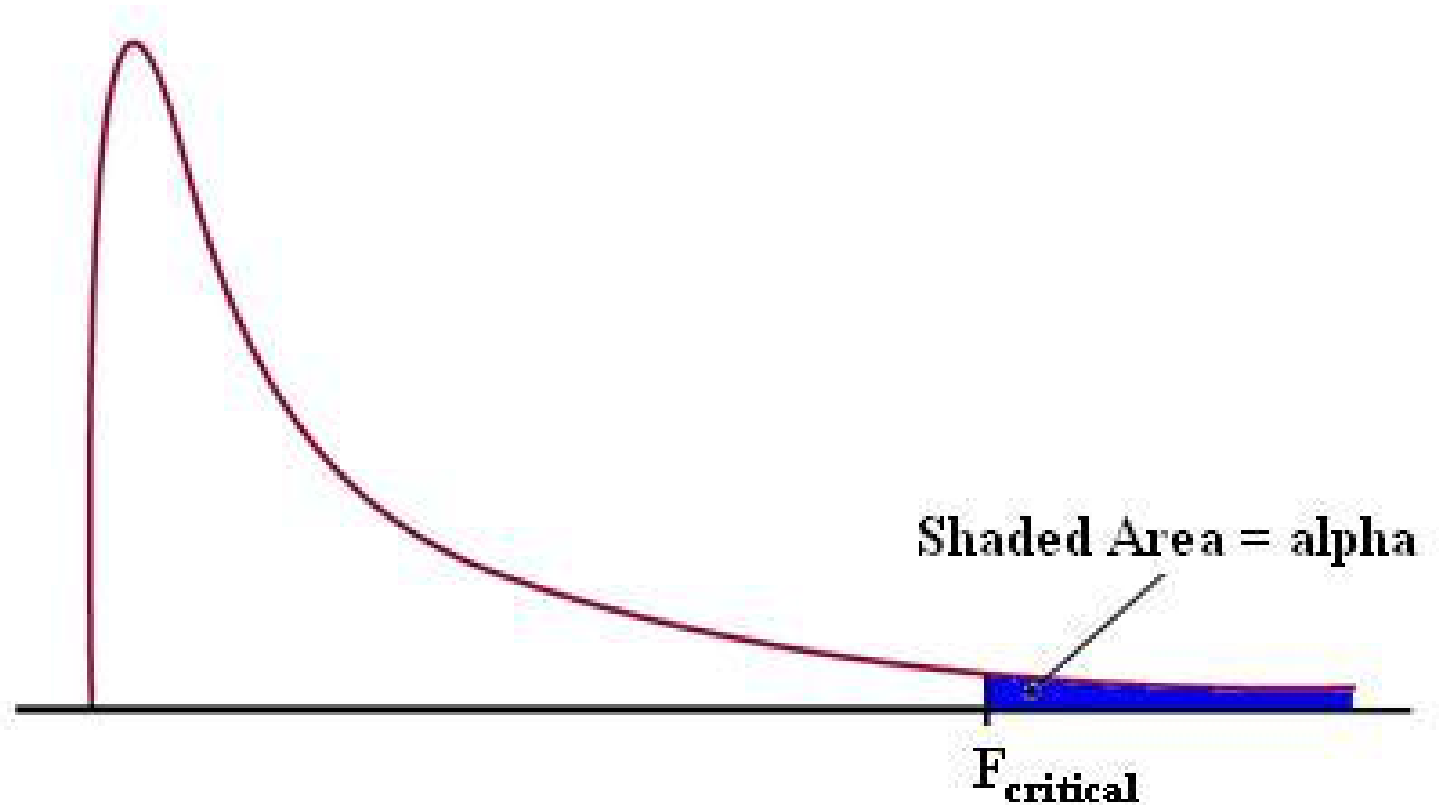


Putting the Pieces together

- The numerator of F is the mean square for groups:
- $MSG = \frac{n_1(\bar{x}_1 - \bar{x})^2 + n_2(\bar{x}_2 - \bar{x})^2 + \dots + n_I(\bar{x}_I - \bar{x})^2}{I - 1}$
- The denominator of F is the mean square for error:
- $MSE = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 + \dots + (n_I - 1)s_I^2}{N - I}$
- When H_0 is true, F has the F distribution with $I - 1$ and $N - I$ degrees of freedom.

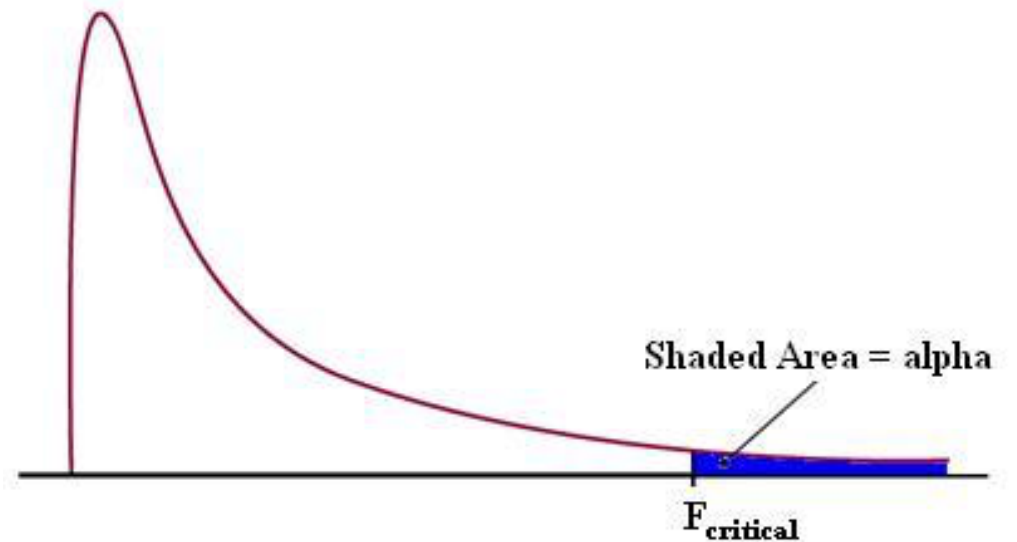
ANOVA and the F statistic

- Therefore, there is only one critical region, in the right tail (shown as the blue shaded region above).
- If the F-statistic lands in the critical region,
 - we can conclude that the means are significantly different, and
 - we reject the null hypothesis.



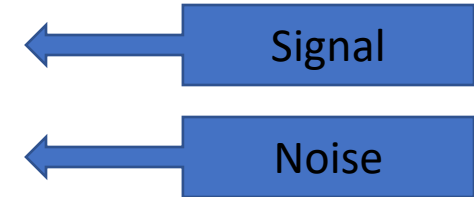
ANOVA and the F statistic

- Again, we have to find the critical value to determine the cut-off for the critical region.
- We'll use the F-table for this purpose.
- We need to look at different F-values for each alpha/significance level
 - because the F-critical value is a function of two things:
 - df_1 and df_2



The F-test

$$F = \frac{\text{Between Groups Variance}}{\text{Within Groups Variance}}$$



$$F = \frac{\text{effect of IV} + \text{individual differences} + \text{experimenter error}}{\text{individual differences} + \text{experimenter error}}$$

If null is true:

$$F = \frac{\text{effect of IV} + \text{individual differences} + \text{experimenter error}}{\text{individual differences} + \text{experimenter error}}$$

What will F = ?

close to 1

Why use ANOVA?

.05!

.05

.05

.05

.05

.05

.05

.05

.05

Reality vs. Behavior
Behavior vs. Psychoanalysis
Psychoanalysis vs. Gestalt
Gestalt vs. Control
Behavior vs. Gestalt
Behavior vs. Control
Reality vs. Gestalt
Reality vs. Control

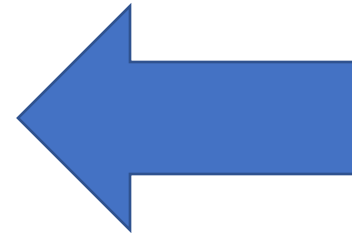
...

If null can be rejected:

$$F = \frac{\text{Big \#} \text{ effect of IV} + \text{individual differences} + \text{experimenter error}}{\text{individual differences} + \text{experimenter error}}$$

What will F = ?

BIG #!



The larger the calculated F, the more likely it is that the difference among condition means reflect true effects of IV rather than error.

ANOVA can tell:

Is there a significant Main Effect of IV1?

Is there a significant Main Effect of IV2?

Is there an interaction?

But where are those differences?

Must do Planned Comparisons
or Post-Hocs to find out where.



	Sm	Med	Lg
Phonics	80	80	80
Whole Word	80	70	60

Wrapping
Up...

